



NAME:

SECTION: HU ID:

LINEAR ALGEBRA

SPRING 2026

QUIZ 9

SECTIONS L1, L2

Max Marks: 10

Time: 10 minutes

Q. Determine whether the following vectors span the vector space R^3 :

$$\mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (2, 3, 4), \mathbf{v}_3 = (1, 3, 4)$$



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Q. Determine whether the following vectors span the vector space R^3 :

$$\mathbf{v}_1 = (0, 0, 1), \mathbf{v}_2 = (1, 2, 3), \mathbf{v}_3 = (2, 4, 6)$$



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SOLUTION QUIZ 9 (L1, L2)

Solution

We must determine whether an arbitrary vector $\mathbf{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$ can be expressed as a linear combination

$$\mathbf{b} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3$$

of the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$. Expressing this equation in terms of components gives

$$(b_1, b_2, b_3) = k_1(1, 1, 2) + k_2(1, 0, 1) + k_3(2, 1, 3)$$

or

$$(b_1, b_2, b_3) = (k_1 + k_2 + 2k_3, k_1 + k_3, 2k_1 + k_2 + 3k_3)$$

or

$$k_1 + k_2 + 2k_3 = b_1$$

$$k_1 + k_3 = b_2$$

$$2k_1 + k_2 + 3k_3 = b_3$$

The problem thus reduces to determining whether this system is consistent for all values of b_1 , b_2 , and b_3 . By parts (e) and (g) of Theorem 4.3.4, this system is consistent for all b_1 , b_2 , and b_3 if and only if the coefficient matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

has a nonzero determinant. However, $\det(A) = 0$ (verify), so $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ do not span $\mathbb{R}^3$.

MARKING SCHEME

[2] + [1] marks for writing a general vector in terms of the given vectors

[3] marks for setting up the system of equations in terms of k 's and b 's

[3] marks for using any method to find out that the system is inconsistent (e.g. that $\det(A) = 0$ using determinant properties, or cofactor expansion)

[1] for concluding that the set does not span the vector space



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QUIZ 9

SECTIONS L3, L4, L5

Max Marks: 10

Time: 10 minutes

Q. (a) Give the definition of a *linearly independent* set of vectors. [3]

(b) Consider $\{v_1, v_2, v_3\}$ which is a linearly dependent set of vectors in a vector space V . If v_4 is any vector in V , then check whether the set $\{v_1, v_2, v_3, v_4\}$ is linearly independent or dependent. [7]



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SOLUTION QUIZ 9 (L3, L4, L5)

(a)

If $S = \{v_1, v_2, \dots, v_r\}$ is a nonempty set of vectors, then the vector equation

$$k_1 v_1 + k_2 v_2 + \dots + k_r v_r = \mathbf{0}$$

has at least one solution, namely

$$k_1 = 0, \quad k_2 = 0, \dots, \quad k_r = 0$$

If this is the only solution, then S is called a **linearly independent** set. If there are other solutions, then S is called a **linearly dependent** set.

(b) This is a simpler version of Q13 from the homework:

To prove this, start from the definition of linear dependence.

Given:

$\{v_1, v_2, v_3\}$ is linearly dependent.

So there exist scalars a_1, a_2, a_3 , **not all zero**, such that:

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = \mathbf{0}$$

Now consider the larger set:

$\{v_1, v_2, v_3, v_4\}$

We want to show this set is also linearly dependent.

Proof:

Take the same equation and just include v_4 with coefficient 0:

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + 0 \cdot v_4 = \mathbf{0}$$

- The scalars $a_1, a_2, a_3, 0$ are **not all zero** (since at least one of a_1, a_2, a_3 is nonzero).
- This is a **nontrivial linear combination** that equals the zero vector.

Conclusion:

$\{v_1, v_2, v_3, v_4\}$ is linearly dependent.

MARKING SCHEME:

Part b

[2] marks for setting up the equation using the definition of linear independence.

[2] marks for writing the new equation with the vector added (with a coefficient of 0)

[2] marks for the reasoning that at least one of the scalars is nonzero, which means the solution is nontrivial.

[1] mark for correct conclusion that the new set is linearly dependent.